

Urban Sprawl Prediction in the Mérida Metropolitan Area (2025–2030)

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Abstract: Rapid urban expansion in the Mérida Metropolitan Area (ZMM) poses a critical threat to the region's karstic aquifer and unique cenote systems. This study develops a hybrid geospatial model combining Cellular Automata (CA) and the LightGBM classifier to project urban growth for 2025–2030. By integrating hydrogeological variables and socio-spatial data, the model demonstrates that AI-based management can optimize urban distribution, significantly reducing fragmentation and protecting vital ecological zones compared to unplanned growth.

Introduction and Objectives

THE PROBLEM OF "INVESTMENT LOTS"



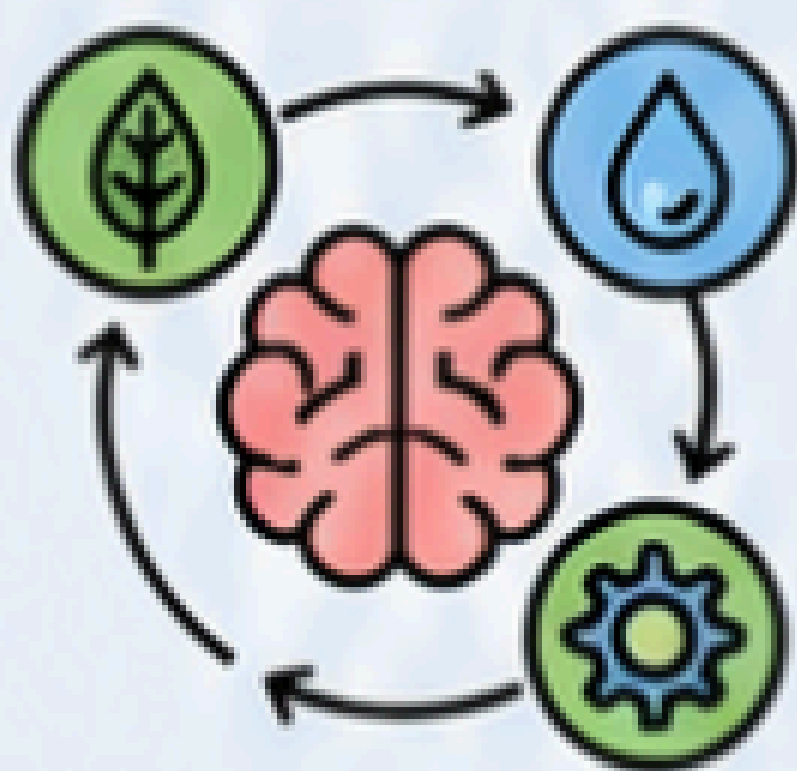
Unregulated peripheral expansion that duplicates the urban footprint without planning the provision of basic services.

RISK TO THE KARSTIC AQUIFER



The soil in Yucatán is highly permeable; urban pollutants seep directly into groundwater, with no natural barriers.

GENERAL OBJECTIVE



Develop a hybrid model CA-LightGBM that integrates socio-spatial and karstic variables to project sustainable scenarios by 2030.

Methodology: The 4-Phase AI Pipeline



Phase 1 and 2: Preparation and Data

Preprocessing in Google Earth Engine (NDVI/NDBI) and construction of the database with 10 spatial variables.



Critical Variables (LightGBM)

Karst Factors: Proximity to cenotes and DRASTIC index (High, Exclusion in AI)

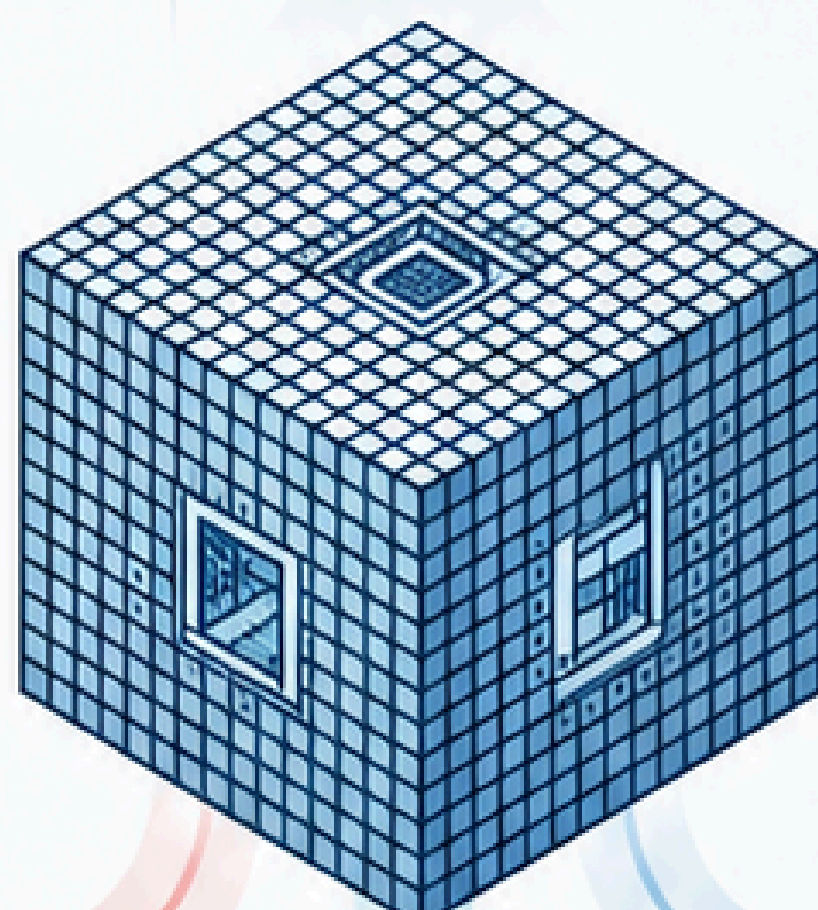
LST (Temperature): Land surface temperature using thermal band (Penalty: heat islands)

Local Density: Urban proximity in 3x3 and 9x9 windows (Main predictor)



Phase 3: Training with LightGBM

Use of gradient boosting classifier with stratified probabilistic sampling to correct data imbalances.



Phase 4: Iterative Simulation (CA)

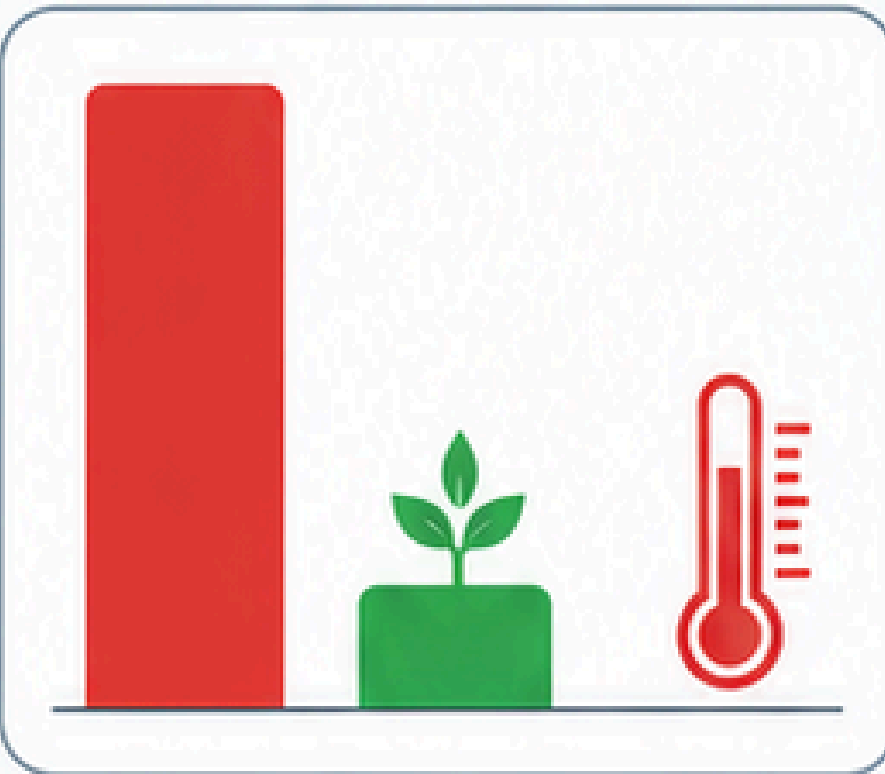
Simulation by Cellular Automata (2024–2030) applying Moore neighborhood rules (3x3).

AI Scenario Optimization:

Achieves the lowest fragmentation (0.20) and protects aquifer recharge zones.

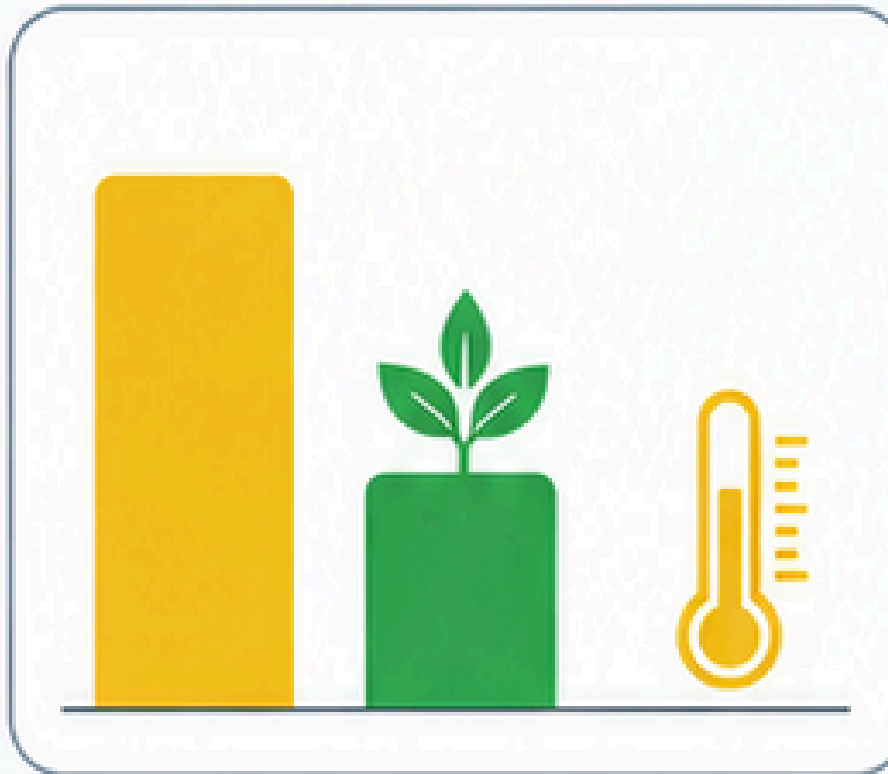
Projection Scenarios for 2030

No Planning (Inertial Growth)



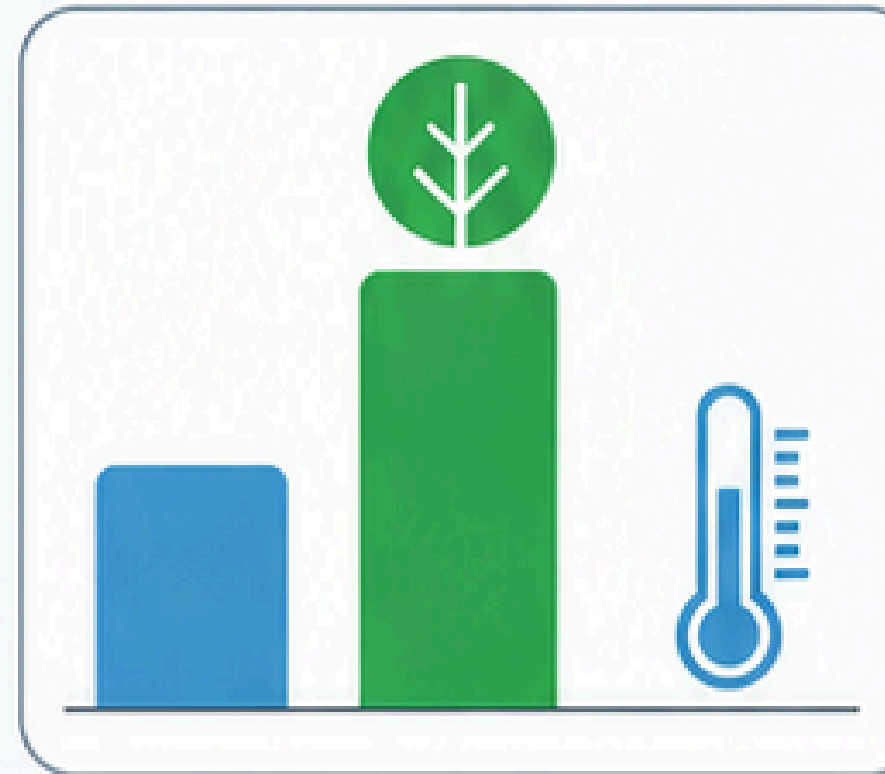
- Urban Area: 1,017 km²
- Green Cover: 5%
- LST Anomaly: +3.6°C

Traditional Plan



- Urban Area: 958 km²
- Green Cover: 23%
- LST Anomaly: +0.4°C

AI-Driven Management



- Urban Area: 911 km²
- Green Cover: 29%
- LST Anomaly: -1.1°C

THERMAL MITIGATION IN AI SCENARIO



CURRENT TREND -1.1°C

Smart management achieves a temperature anomaly of -1.1°C compared to the current trend, helping to combat the heat island effect.

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RESULTS AND DISCUSSION

Comparative Growth Scenarios For 2030

Analytical Indicator	No Planning (Inertial)	Traditional Plan	AI Management (Optimal)
Total Urban Area	1,017 km ²	958 km ²	911 km ²
Fragmentation Index	0.67 (High)	0.26 (Low)	0.20 (Minimal)
Green Cover %	5%	23%	28%
LST Temperature Anomaly	+3.6°C	+0.4°C	-1.1°C
Aquifer Vulnerability	0.68 (Critical)	0.36 (Moderate)	0.24 (Controlled)

DISCUSSION AND IMPLICATIONS

- IA Management saves \$106 km² of expansion.
- Distance to cenotes: 2nd most important predictor, confirming forgotten karst vulnerability.
- Aquifer vulnerability controlled (\$0.24 vs 0.68 critical).
- Temperature anomaly reversed (–1.1°C vs +3.6°C critical).

AI Management (Optimal)



Traditional Plan



No Planning (Inertial)



CONCLUSION

- The hybrid LightGBM-CA model is a powerful tool for simulating complex urban dynamics in fragile ecosystems.
- Incorporating specialized karstic variables significantly improves the realism of growth simulations.
- AI-driven optimization successfully channelling development away from high-vulnerability zones, mitigating the urban heat island effect, and preserving vital green corridors.

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MAIN CONCLUSION
AI enables multi-objective optimization that saves up to 106 km² of unnecessary expansion compared to traditional planning